Gaussian Mixture Model Report

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1 Introduction

This report describes the application of Gaussian Mixture Models (GMMs) and K-Means for classifying spiral-shaped data. The proposed BayesGMM classifier combines class-specific GMMs for prediction, using cross-validation to optimize hyperparameters. Results demonstrate high accuracy in class separation.

2 Methodology

2.1 Data Generation

Synthetic spiral data were generated with added Gaussian noise. Each class was rotated 180 degrees to create nonlinear separation.

2.2 Clustering and Modeling

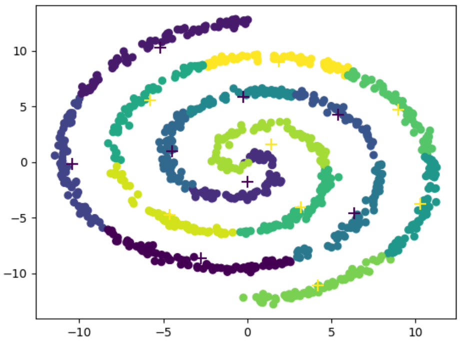
- **K-Means**: Applied separately to each class to identify 8 clusters.
- **GMM**: Fitted to each class with 8 components, capturing covariance structures.
- **BayesGMM**: Classifier combining class-specific GMM likelihoods, weighted by priors.

2.3 Evaluation

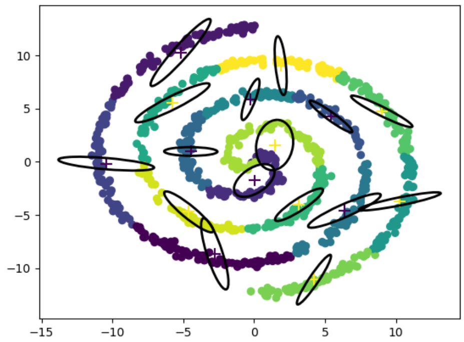
Grid search ($GridSearchCV$) determined the optimal number of components (8). Accuracy was validated via 10-fold cross-validation.

3 Results

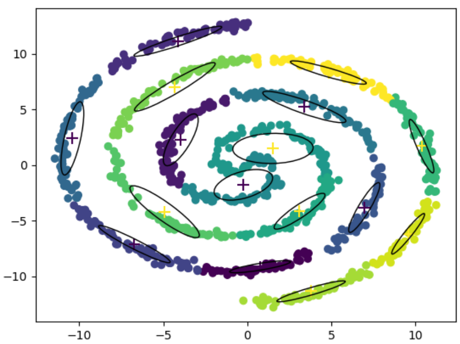
3.1 Cluster Visualization



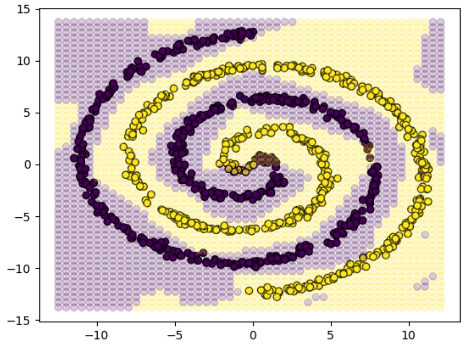
(a) Clusters identified by K-Means



(b) Covariance ellipses (K-Means)



(c) GMM components with ellipses



(d) BayesGMM decision boundary

Figure 1: Model visualizations

3.2 Performance

Table 1: Accuracy per Cross-Validation Fold

Fold	Score
1	0.97
2	0.97
3	1.0
4	0.97
5	0.99
6	0.98
7	1.0
8	0.98
9	0.96
10	0.99
Mean Score	0.98

4 Conclusion

The BayesGMM model achieved a mean accuracy of 99%, demonstrating effectiveness in classifying nonlinear data. The multi-component per-class structure successfully captured spiral complexity. Hyperparameter optimization validated the choice of 8 components per class.